

Date: 26 October, 2025

- **Kleene's Theorem**

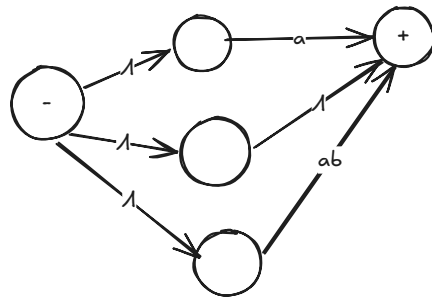
- A language that can be defined using regular expression, can also be expressed using finite automata, and can also be expressed in transition graph.
- The theorem has 3 parts
 - A language that can be defined by finite automata can also be defined by transition graph
 - A language that can be defined by transition graph can also be defined with regular expression.
 - A language that can be defined by regular expression can also be defined by finite automata.
 - $RE \quad FA \quad TG \quad RE$

- **Proof**

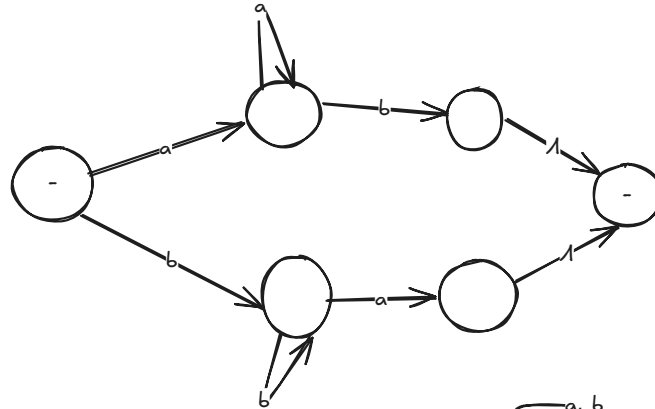
- By definition, every FA is also a TG. (1st Part is proven by definition)
- Second Part requires constructive proof.
 1. Label states.
 2. Make multiple starting states as single state.
 3. Make multiple final steps into a single final state.
 4. If a TG has same start and final state, separate them.
 5. Get ride of intermediary states by bypassing them and converting them into their respective RE.

TG Proof

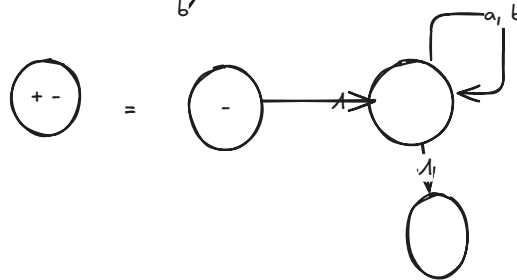
1.



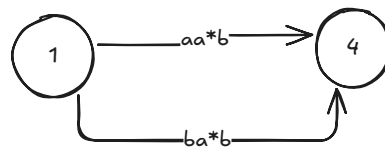
2.



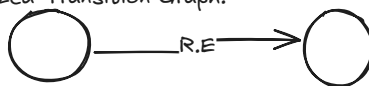
3.



4. We can bypass states in TG



5. Generalized Transition Graph.



More edges from same state to same state